

A Gaussian-rational degree-two counterexample to convex closure of exterior convex maps

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Abstract

We give two Laurent polynomials of the form $z + a/z + b/z^2$, with $a, b \in \mathbb{Q}(i)$, that are univalent on $|z| > 1$ and map that domain onto complements of nonempty compact convex sets. Their midpoint is also univalent but does not have a convex omitted set. This answers affirmatively the existence question within the degree-two finite-tail class. The endpoint properties follow from elementary injectivity bounds and two quartic sum-of-squares identities. Failure at the midpoint is certified by an exact negative rational value of the exterior-convexity criterion at $z = 100000/99999$.

1 Statement and context

Write $\mathbb{D}^* = \{z \in \mathbb{C} : |z| > 1\}$. We call a normalized holomorphic map on $\mathbb{D}^*$ *exterior convex* if it is injective and its image is $\mathbb{C} \setminus K$ for a nonempty compact convex set K . The degree-two existence question asks whether two exterior convex maps $z + a/z + b/z^2$ with Gaussian-rational coefficients can have a rational convex combination that is not exterior convex. The following explicit witness answers this question.

Theorem 1. *Set*

$$F(z) = z + \frac{-5 + 17i}{20z} + \frac{-1075 + 731i}{36560z^2}, \quad (1)$$

$$G(z) = z + \frac{-3 + 15i}{20z} + \frac{-1909 + 1245i}{30160z^2}. \quad (2)$$

Both F and G are exterior convex. Their midpoint

$$H(z) = \frac{F(z) + G(z)}{2} = z + \frac{-1 + 4i}{5z} + \frac{-159711 + 105569i}{3445780z^2} \quad (3)$$

is injective on $\mathbb{D}^$, but its omitted compact set is not convex.*

Guo and Hu [1] give a counterexample to the general convex-closure problem for normalized exterior convex functions. One map in their construction is defined by a fractional-power derivative and has an infinite Laurent tail; their equation (17) displays a nonzero z^{-5} term. The contribution here is an exact witness entirely within the $z + a/z + b/z^2$ class. No truncation argument is used to preserve convexity. The comparison is limited to the inspected sources and does not assert exhaustive priority.

For adjacent use of complex-variable reductions and exact rational certificates in spectral inequalities, see Zeng [2]. That work provides methodological context, not a result about convex closure of Laurent maps.

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2 Proof of Theorem 1

Injectivity and the omitted sets

For $f(z) = z + a/z + b/z^2$, define

$$D(w) = 1 - aw^2 - 2bw^3, \quad N(w) = 1 + aw^2 + 4bw^3. \quad (4)$$

Then, with $w = 1/z$,

$$f'(z) = D(w), \quad 1 + \frac{zf''(z)}{f'(z)} = \frac{N(w)}{D(w)}. \quad (5)$$

The condition $|a| + 2|b| < 1$ implies that D has no zero on $|w| \leq 1$. It also proves global injectivity on the closed exterior: for distinct z, v with $|z|, |v| \geq 1$,

$$\frac{f(z) - f(v)}{z - v} = 1 - \frac{a}{zv} - \frac{b(z+v)}{z^2v^2}, \quad (6)$$

and the modulus of the terms subtracted from 1 is at most $|a| + 2|b| < 1$.

Denote the coefficients of F, G, H by (a_F, b_F) , (a_G, b_G) and (a_H, b_H) . Direct squared-modulus calculations give

$$\begin{aligned} |a_F|^2 &= \frac{157}{200} < \left(\frac{9}{10}\right)^2, & |b_F|^2 &= \frac{1849}{1462400} < \left(\frac{1}{25}\right)^2, \\ |a_G|^2 &= \frac{117}{200} < \left(\frac{4}{5}\right)^2, & |b_G|^2 &= \frac{6889}{1206400} < \left(\frac{2}{25}\right)^2. \end{aligned}$$

Consequently

$$|a_F| + 2|b_F| < \frac{49}{50}, \quad |a_G| + 2|b_G| < \frac{24}{25}, \quad |a_H| + 2|b_H| < \frac{97}{100}, \quad (7)$$

where the last bound follows by averaging and the triangle inequality. Thus all three maps are injective on the closed exterior and have nonvanishing derivatives there.

For each map the image of the unit circle is a regular Jordan curve. Its open exterior image is the unbounded component of the complement of this curve. To see the surjectivity onto that component, the image is open, connected, disjoint from the curve, and contains points tending to infinity because $f(z) = z + O(z^{-1})$. It is also relatively closed in that component: a convergent image sequence has bounded preimages, and a preimage limit cannot lie on the unit circle when the image limit is off the curve. Connectedness of the component completes the argument. The omitted set is therefore the closure of the bounded Jordan component, and is nonempty and compact. It remains to decide its convexity.

Endpoint convexity for every boundary angle

We use the standard exterior-convexity criterion, in the form stated in [1, equation (2)]: for a normalized locally univalent map on $\mathbb{D}^*$, the exterior-convexity property is characterized by

$$\operatorname{Re} \left(1 + \frac{zf''(z)}{f'(z)} \right) > 0 \quad (z \in \mathbb{D}^*). \quad (8)$$

Our maps already satisfy global injectivity and the compact-complement condition, as just proved.

For $w = e^{-it}$, multiplication in (4) yields

$$\begin{aligned} Q_{a,b}(t) &:= \operatorname{Re}(N(w)\overline{D(w)}) \\ &= 1 - |a|^2 - 8|b|^2 + 2\operatorname{Re}(be^{-3it}) - 6\operatorname{Re}(a\bar{b}e^{it}). \end{aligned} \quad (9)$$

Equivalently, for $\gamma(t) = f(e^{it})$, this quantity is $\text{Im}(\overline{\gamma'(t)}\gamma''(t))$. Put $T = \tan(t/2)$, which covers the unit circle except $t = \pi$ modulo 2π . Substitution of the coefficients in (1)–(2) gives

$$Q_F(t) = \frac{43T^2 P_F(T)}{91400(1+T^2)^3}, \quad P_F(T) = 871T^4 - 258T^2 - 2720T + 2871, \quad (10)$$

$$Q_G(t) = \frac{83T^2 P_G(T)}{75400(1+T^2)^3}, \quad P_G(T) = 671T^4 - 498T^2 - 2400T + 2511. \quad (11)$$

These identities can also be checked directly by substituting $w = (1 - T^2 - 2iT)/(1 + T^2)$ into (4). The two quartics have the exact decompositions

$$P_F(T) = 871(T^2 - 1)^2 + \frac{4}{371}(371T - 340)^2 + \frac{279600}{371}, \quad (12)$$

$$P_G(T) = 671(T^2 - 1)^2 + \frac{4}{211}(211T - 300)^2 + \frac{28240}{211}. \quad (13)$$

All square weights and constant terms are positive. Hence $P_F, P_G > 0$ on $\mathbb{R}$, and $Q_F, Q_G \geq 0$ at all angles covered by T . The missing angle satisfies

$$Q_F(\pi) = \frac{37453}{91400} > 0, \quad Q_G(\pi) = \frac{55693}{75400} > 0. \quad (14)$$

Thus both boundary functions are nonnegative on the full circle, with equality only at $t = 0$ modulo 2π .

For either endpoint, N/D is holomorphic on a neighborhood of the closed unit disk by (7). Its real part is harmonic there, has boundary values $Q_{a,b}/|D|^2 \geq 0$, and equals 1 at $w = 0$. The minimum principle gives nonnegativity in the disk; the strong minimum principle excludes any interior zero. Therefore $\text{Re}(N(w)/D(w)) > 0$ for $|w| < 1$. Equations (5) and (8) now prove that F and G are exterior convex. Their isolated zero boundary curvature does not violate the strict criterion in the open domain.

An exact interior violation for the midpoint

For H in (3), let D_H, N_H be as in (4) and set

$$w_0 = \frac{99999}{100000}, \quad z_0 = \frac{100000}{99999} > 1.$$

Direct rational arithmetic gives

$$\begin{aligned} \text{Re}(N_H(w_0)\overline{D_H(w_0)}) &= -\frac{135836899642077974393353880656369}{4307225000000000000000000000000000}, \\ |D_H(w_0)|^2 &= \frac{20785093282881895304598732129781306369}{8614450000000000000000000000000000}. \end{aligned}$$

In particular,

$$\text{Re}\left(1 + \frac{z_0 H''(z_0)}{H'(z_0)}\right) = -\frac{271673799284155948786707761312738}{20785093282881895304598732129781306369} < 0. \quad (15)$$

Necessity in (8) excludes a convex omitted set for H . For comparison, its boundary numerator already has the negative value $Q_H(0) = -63/861445$, but (15) gives a violation strictly inside the exterior domain. This completes the proof of Theorem 1. $\square$

3 Exact computation and scope

The two sum-of-squares identities prove positivity for every real T ; they are not observations on a finite angular mesh. The separate check at $t = \pi$ completes the boundary coverage. The midpoint calculation uses exact rational arithmetic at one specified point, which suffices to refute the universal condition (8). No assertion about the full degree-two coefficient region is needed.

The accompanying files `verify.py` and `witness.json` replay the coefficient arithmetic using only Python's standard library, including `fractions.Fraction`. Run `python verify.py` from their directory. The verifier substitutes the rational circle parameter into N, D , checks the complete polynomial identities, the positive square weights and constants, the omitted angle, the tail bounds, the midpoint coefficients, and the exact interior negative value. It prints its results without modifying the inputs. The analytic criterion and the harmonic minimum principle are proved or invoked in the text, not formally encoded by this finite calculation.

Computational and AI assistance. AI assistance was used in exploring boundary-curvature expressions, finding and rationalizing the coefficients, deriving the displayed identities, preparing verification code, and drafting this article. Floating-point exploration was followed by exact rational verification. A separate AI review checked the analytic argument and recomputed the finite certificates by direct real-coordinate differentiation in SymPy. These checks are not formal proof-assistant certification or external peer review; the proof is the argument and identities given above.

References

- [1] Y. Guo and X. Hu, *A Counterexample to a Problem of Pommerenke on Convex Functions in the Class Σ* , arXiv:2609.04279v1 (2026). <https://arxiv.org/abs/2609.04279v1>.
- [2] Z. Zeng, *Fixed Convex-Lens Spectral Constants: Möbius Reduction, Sharp Model Theorems, and Angle-Dependent Bounds*, Preprints (2026). <https://doi.org/10.20944/preprints202608.1272.v1>.